

Confidential

Long-Term 2029: Stability Feasibility Requirements

Technical Feasibility Study

Tender Year: 2025–2026

Delivery Year: 2029–2039

Confidential

Disclaimer

Information that is supplied in this document as part of this ITT is supplied in good faith. The information contained in the Long-term 2029 ITT Pack and in any related written or oral communication is believed to be correct at the time of issue, but the National Energy System Operator (NESO) will not accept any liability for its accuracy, adequacy or completeness and no warranty is given as such. Any reliance by a Tenderer or any third party on the information provided by NESO as part of this EOI Pack is at the Tenderer's and/or third party's risk.

Version Control

Version Number	Date	Notes
V1	16 July 2025	Initial version published at ITT.
V2	22 August 2025	This version includes updates to Frequency events and Voltage angle change events. Updates are marked with a 'V2' indicator.
V3	10 September 2025	Clarification for solutions part of a larger installation. Updates are marked with a 'V3' indicator.
V4	10 October 2025	A small error in Test 2 was corrected. Updates are marked with a 'V4' indicator
V5	18 November 2025	Clarification added to Test 2 for 0MW generators. Updates are marked with a 'V5' indicator.
V6	05 March 2026	Updated version to clarify feasibility simulation tests and expected outcomes of tests. Updates are marked with a 'V6' indicator.

Confidential

Contents

Disclaimer	2
Version Control	2
Overview	4
1. Purpose of this document.....	4
2. V6: How simulations will be used.....	4
3. Scope of simulation tests required	4
4. Outputs	5
5. Technical queries	6
6. Changes later in process	6
Test 1. Short-circuit events	9
Test 2. Frequency events.....	13
Test 3. Voltage angle change events	22

Confidential

Overview

1. Purpose of this document

V6: This section has been clarified as part of the V6 update.

The purpose of this document is to set out the desktop-based feasibility simulation studies that are required as part of the Long-term 2029 tender in relation to the Stability Service Technical Submission.

The simulation test results submitted by bidders, and any supporting information provided, must be based on factual statements relevant to the Long-term 2029 Stability Technical Specification with relevant references and desktop-based simulation outcomes including:

- description of the proposed solution and its key technical considerations relevant to the technical specification.
- any specific data needs that would be critical to the proposed solution.
- feasibility study demonstration for each proposed solution.

2. V6: How simulations will be used

V6: This section has been clarified as part of the V6 update.

The stability feasibility study submitted by bidders containing the outcomes of the simulations allows NESO to understand the key technical capabilities of proposed solutions in relation to the Stability Technical Specification.

Ultimately, submitted simulation results will be reviewed when assessing submitted Stability Technical Proformas, particularly when scoring the questions within tab A Solution Outline and tab B Feasibility Simulation.

3. Scope of simulation tests required

V6 This section has been clarified as part of the V6 update.

Table 1 sets out the key performance criteria of the technical specification that will need to be demonstrated as part of this technical study. Specific simulation tests that are required are set out in Appendix A.

Please note that any simulation tests carried out and capabilities demonstrated during the Long Term 2029 tender do not remove proving or compliance testing requirements before and after commissioning of the solutions.

Confidential

Table 1: List of tests with reference to ECC.6.3.19

Description	Feasibility Test (Appendix A references) and notes
Short circuit current value	Test 1 – value determined at the technical study will be used in the economic assessment & contract
Inertia value	Test 2 – value determined at the technical study will be used in the economic assessment & contract
Response within 5 ms	Tests 2, 3
Inertia behaviour	Test 2
V6 Voltage source behind an impedance	Tests 2, 3
Phase Angle jump response	Test 3

4. Outputs

The required output is a technical feasibility study report covering all solutions submitted by a Tenderer during the tender. A report template will be provided at the ITT stage of the tender, which Tenderers **must** complete in conjunction with the Stability Technical Proforma.

- Each Tenderer must provide a complete technical study report. The technical study report must be completed using the NESO template provided and must include the simulation results of all the relevant tests in Appendix A for each solution that has been submitted.
- The report must demonstrate compliance with the technical specification as described in this document.
- The report must be in Plain English. Where the report relies on data to demonstrate compliance, the data must be shown in the report in the form of a graph or figure that shall be clearly legible including any axes or legends.
- Where the report relies on equipment specification, copies of manufacturer documentation must be attached to the report as appendices. Where possible, bidders should sign post what pages of appendix documentation that NESO should refer to.

Confidential

- Where the demonstration of compliance is ambiguous in the report provided, NESO may seek additional clarification and request additional information including but not limited to raw data, models, and additional study results. For avoidance of doubt this will not be an opportunity for a different or new submission, it will purely be a clarification request for additional information. For more details on clarifications, refer to the Contract Award Criteria.
- The report must show the capability of the solution to respond to both frequency events (Up to 1 Hz/s RoCoF events only for this report) and voltage angle change events.

5. Technical queries

Any technical queries must be submitted to NESO in accordance with the query process outlined in the Instructions to Tenderers document.

6. Changes later in process

Information provided in the technical feasibility study at the time of tender submissions will be used in the tender assessment and cannot be changed by bidders post tender submission.

V6: NESO reserve the right to clarify submissions in line with the LT2029 Contract Award Criteria.

Confidential

Appendix A: List of desktop simulations

This appendix provides a list of desktop simulations that are required as part of the technical study. For each test category, the Tenderer must give an overview of the test method and provide output results, observations, limitations in clear English and in a legible format.

All tests are required for each solution submitted, for each technology type, rating, and substation location within the solutions.

Grid Entry Point (GEP)

Unless otherwise stated in individual tests, all feasibility test results must be shown at the Grid Entry Point. The Grid Entry Point is defined as the point where the solution is directly or radially connected to the transmission system. Any equipment between the solution and the Grid Entry Point that impact the solution's performance (e.g. connection transformers/cables/circuits etc.) must be explicitly modelled.

The test simulation must be run for long enough to allow the system to settle to a steady state before any event is applied and long enough after the test to allow the system to return to steady state.

Unless otherwise stated:

- all positive sequence RMS results must be recorded.
- for EMT simulation, EMT and RMS quantities must be provided. The method adopted to compute RMS values must be clearly stated and explained.
- For all simulations a suitable time-step must be used, for GBGF-I solutions please refer to the [GBGF best practice guide](#)
- The time constant for EMT to RMS conversion (for GBGF-I and Hybrid solutions), must be 20ms.
- All results must be recorded with step sizes not greater than 1ms.
- Both AC and DC components must be included for EMT measured current.

For every test, the following must be recorded:

- voltage magnitude and phase angle at the Grid Entry Point and solution terminal.
- active power and reactive power at the Grid Entry Point and solution terminal.
- active, reactive, and total current at the Grid Entry Point and solution terminal.
- frequency and RoCoF at the Grid Entry Point.
- Tap Changer Positions for any transformers for the duration of the test.

Confidential

An Excel workbook or a csv file for **every test**, containing the above recorded measurements, must be submitted as a supplementary dataset along the feasibility study report. For Hybrid solutions, datasets of the measurements at the GBGF-I terminals must be provided in addition to the datasets of the measurements at the GEP for the overall hybrid solution.

V2 The solution terminal is the first AC point where the Grid Forming Capability can be demonstrated. For a GBGF-S solution this would be the terminals of the machine, for a GBGF-I solution this would be the AC side of the inverter. For plant which may be made up of the multiple machines, all solution terminals should be provided. For GBGF-I plant which utilise a lumped model of multiple inverters, the solution terminal can be provided at the terminal of the lumped model.

The test model must be set up as follows:

- a. The solution must be modelled in EMT for GBGF-I and Hybrid solutions. For GBGF-S solutions RMS models are required. The model in all cases must accurately reflect the actual solution's performance and limitations.
- b. Any equipment that impacts the performance at the Grid Entry Point must be modelled.
- c. The total system equivalent must be modelled as an ideal voltage source behind an impedance.
- d. Nominal settings and ratings of assets must be used in the model and simulations. When a range of parameters is indicated in the manufacturer datasheet, the simulations can be performed considering the nominal values.
- e. All simulation settings, model parameters, transformer tap changer position, model settings and controllers' parameters must remain unchanged for all tests. Any change in these settings and parameters, other than those requested by NESO (e.g. reactive power set point, frequency of the system, etc.) must be declared, justified and approved by NESO.
- f. **V3** If the solution tendered for the stability service is part of a wider installation that shares a common Grid Entry Point, then the impact of the other plant and apparatus that has an impact upon the total performance at the GEP will need to be included as a part of the feasibility study.

The results from this plant must be included in the data submission. This must cover scenarios where the other plant (Plant that shares the GEP but is not entering the tender) is in service and out of service demonstrating the impact that this has on the service provision from the machine tendered. For clarity, any contribution this plant makes to the GEP must be removed from the contribution from the plant for Inertia and Short-Circuit Level injection quantities.

Confidential

Test 1. Short-circuit events

V6 The Outcome of these tests must demonstrate:

- the fault current injection at the Grid Entry Point during a fault at the Grid Entry Point.
- the fault current injection at the Grid Entry Point in response to a retained voltage experienced at the Grid Entry Point as a result of a remote fault at a specified Reference point.
- for GBGF-I, that the fault current injection at the Grid Entry Point starts within 5ms and reaches the Peak Current Rating no later than 30ms of fault occurrence for zero retained voltage at Grid Entry Point.

Failure to demonstrate these requirements may result in the solution failing the Stability Technical Criteria.

1.1 Test conditions

For all solutions, please refer to Table 2 for a description of the tests required. Please note that Steps 1-4 can be omitted for solutions that do not have the capability to export or import MW. However, for numbering consistency, please keep numbering the steps as Steps 5-8 even if Steps 1-4 are not performed.

Steps 1-7 are required to establish the lowest fault current for a range of credible operating conditions. Step 8 is required to understand the fault current contribution of the solutions with respect to specific Reference point(s), which are equivalent to the points on the network where the stability services are required. As the fault current contribution to the Reference points will depend on the impedance between the Grid Entry Point and the Reference point, Step 8 will need to be repeated for each reference point within the region(s) being targeted.

Throughout these steps, the short circuit current contribution from the solution must be recorded as the positive sequence RMS fault current at 40ms after the fault initiation, measured in kA at the Grid Entry Point, following a three phase to earth fault at the stated location. This RMS quantity must be calculated using the DQ0 transform method to obtain the RMS value from the EMT simulation where applicable. It is the lowest short circuit current value from Steps 1-7 which will be entered in the service contract against which contract payments will be considered. Effectiveness factors, which have been provided by NESO at the time of tender launch, will be considered to evaluate the fault current contribution from the Grid Entry Point (derived from Step 8) to the relevant Reference points. The results from Step 8 will be considered in the commercial assessment, Step 8 must be repeated for all reference nodes that the solution is effective towards.

All solutions must demonstrate all short circuit values from Test 1 Steps 1-7 in the compliance stage and proving tests along with requested aspects of the technical specification. Service commencement will be dependent on this demonstration.

Table 2 below are for all solutions and present the different steps for the short circuit test along with the network configuration, initial conditions and fault event applicable for each step.

Confidential

For each **solution**, the tests described in Table 2 are required:

Table 2: Test 1: Simulations required for each solution.

Step	Network Configuration	Initial Conditions	Simulated Event
1	Figure 1	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power export (generation). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 3 phase to earth fault at the Grid Entry Point that is cleared after 140ms.
2	Figure 1	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power export (generation). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 3 phase to earth fault at the Grid Entry Point that is cleared after 140ms.
3	Figure 1	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power import (demand). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 3 phase to earth fault at the Grid Entry Point that is cleared after 140ms.
4	Figure 1	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power import (demand).	Simulate a 3 phase to earth fault at the Grid Entry Point that is cleared after 140ms.

Confidential

		Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	
5	Figure 1	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at zero active power output. Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 3 phase to earth fault at the Grid Entry Point that is cleared after 140ms.
6	Figure 1	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at zero active power output. Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 3 phase to earth fault at the Grid Entry Point that is cleared after 140ms.
7	Figure 1	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at zero active power output. Solution operating at zero reactive power output.	Simulate a 3 phase to earth fault at the Grid Entry Point that is cleared after 140ms.
8	Figure 2	From steps 1 to 7, the combined reactive power and active power outputs resulting in the lowest short circuit current injected at the GEP must be used as the active and reactive power setpoints for Step 8. Repeat Step 8 for all specified Reference point(s) for the targeted locations.	Simulate a 3 phase to earth fault at a Reference point on the network, where the fault impedance between the Reference point and the GEP is Z_f . The fault must

Confidential

		be cleared after 140ms.
--	--	-------------------------

Number of simulations for each solution: minimum of 8 per locations targeted for solutions with non-zero MW capability and minimum of 4 (Steps 5-8) for zero MW solutions, assuming that the Grid Entry Point is not the same as the Reference point(s). Please note that some locations may have to simulate faults with respect to multiple Reference points within a region or across multiple regions (e.g. for substations in the North East England, South Wales and West England regions, South West England, North Scotland).

If the solution is planned to connect to a substation where a bay has been reserved for the Stability Long-Term Market, the maximum current from Steps 1-7 must not exceed the reserved short circuit capacity of the bay.

Network Configuration for Short Circuit Test

Figure 1 illustrates the single-line diagram of the system to be simulated in Test 1, Steps 1-7. The equivalent system impedance (Z_{sys}) for each location has been provided by NESO at the time of tender launch.

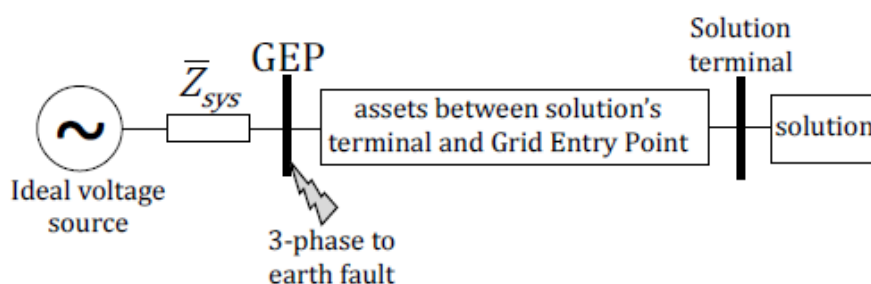


Figure 1: Network configuration for Test 1, Step 1-7

Figure 2 illustrates the single-line diagram of the system to be simulated in Test 1, Step 8. The equivalent system impedance (Z_{sys}) and the fault impedance (Z_f) for each location has been provided by the NESO at the time of tender launch. Please note that for the fault at the Reference point, only **positive (injection) reactive fault** current is considered in the assessment. Any negative reactive fault current (absorption) will be considered as null contribution.

Confidential

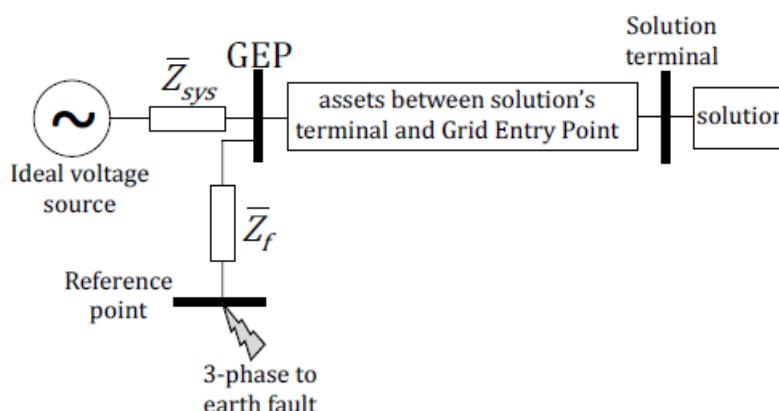


Figure 2: Network configuration for Test 1, Step 8

Test 2. Frequency events

The purpose of this test is to understand the inertial response of the solution.

V6 The Outcome of these tests must demonstrate

- a response to a change in frequency with a change in active power output within 5ms.
- an inertial response equal to the amount to be declared in the tender.
- V6 That the solution meets the relevant requirements set out in Section 4.3 of the Technical Specification.

Failure to demonstrate these requirements may result in the solution failing the Stability Technical Criteria.

The frequency events must be modelled as a change in the grid source frequency. Using other methods such as sudden decrease/increase of demand or generation will not be accepted.

All Frequency Response (LFSM/FSM) must be disabled.

V6 The tests should be completed with voltage control enabled, to maintain the Grid Entry Point Voltage at 1 p.u.

Confidential

2.1 Test conditions

In the following frequency events:

- calculation must show how the performance in the tests relates to the declared values for inertia (MVA.s).¹

V5 For all solutions, please refer to Table 3 for a description of the tests required. For solutions which do not have the ability to import or export active power, only steps 5-12 and 14 are required. However, for numbering consistency, please keep numbering the steps as Steps 5-12 even if other steps are not performed.

For each **solution**, the tests described in Table 3 are required:

Table 3: Test 2: Simulations required for each solution.

Step	Network Configuration	Initial Conditions	Simulated Event
1	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power export (generation). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate frequency event to drop from 50Hz to 49Hz with RoCoF of 1Hz/s.
2	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power export (generation). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate frequency event to drop from 50Hz to 49Hz with RoCoF of 1Hz/s.
3	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u.	Simulate frequency event to increase

¹ Note: We consider 1 GW.s = 1 GVA.s for the provision of inertia.

Confidential

		If applicable, solution operating at maximum active power import (demand). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	from 50Hz to 51Hz with RoCoF of 1Hz/s.
4	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power import (demand). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate frequency event to increase from 50Hz to 51Hz with RoCoF of 1Hz/s.
5	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at zero active power output. Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate frequency event to drop from 50Hz to 49Hz with RoCoF of 1Hz/s.
6	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at zero active power output. Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate frequency event to drop from 50Hz to 49Hz with RoCoF of 1Hz/s.
7	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at zero active power output.	Simulate frequency event to increase from 50Hz to 51Hz with RoCoF of 1Hz/s.

Confidential

		Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	
8	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at zero active power output. Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate frequency event to increase from 50Hz to 51Hz with RoCoF of 1Hz/s.
9	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at zero active power output. Solution operating at zero reactive power output.	V2 Simulate a frequency step event from 50Hz to 49Hz lasts for 2s.
10	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power export (generation). V2 For 0MW solutions this should be completed at 0MW. Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate 5 successive Frequency drops from 50Hz to 49Hz with a RoCoF of 1Hz/s. At the end of the RoCoF event the frequency event must be removed as a step change, the time between the end of one event and the start of the next must be 5 seconds.(An example image of the first two frequency injections is shown in Figure 3)

Confidential

11	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power Import (Demand). V2 For 0MW solutions this should be completed at 0MW. Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate 5 successive Frequency rises from 50Hz to 51Hz with a RoCoF of 1Hz/s. At the end of the RoCoF event the frequency event must be removed as a step change, the time between the end of one event and the start of the next must be 5 seconds. (An example image of the first two frequency injections is shown in Figure 4)
V2 12	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power export (generation). Solution operating at zero reactive power output. V5 This step should be completed for 0MW solutions at 0MW and zero reactive power output.	Simulate a frequency decrease from 50Hz to 47Hz at a rate of 1Hz/s over a 3 second period. Allow conditions to stabilise for 5 seconds and then increase the System Frequency from 47Hz to 52 Hz at a rate of 1 Hz/s over a 5 second period. V4 Allow conditions to stabilise for 5 seconds and then decrease the System Frequency from 52Hz to 50Hz over a 2

Confidential

			second period at a rate of 1 Hz/s.
V2 13	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power import (demand). Solution operating at zero reactive power output. V5 This step does not need to be completed by 0MW solutions.	Simulate a frequency decrease from 50Hz to 47Hz at a rate of 1Hz/s over a 3 second period. Allow conditions to stabilise for 5 seconds and then increase the System Frequency from 47Hz to 52 Hz at a rate of 1 Hz/s over a 5 second period. V4 Allow conditions to stabilise for 5 second and then decrease the System Frequency from 52Hz to 50Hz over a 2 second period at a rate of 1 Hz/s.
V2 14	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power export (generation). Solution operating at zero reactive power output. V5 This step should be completed for 0MW solutions at 0MW and zero reactive power output.	Simulate a frequency increase from 50Hz to 52Hz at a rate of 1 Hz/s over a 2 second period. Allow conditions to stabilise for 5 seconds and then decrease the System Frequency from 52Hz to 47Hz at a rate of 1 Hz/s over a 5 second period.

Confidential

			Allow conditions to stabilise for 5 seconds, then simulate a frequency increase from 47Hz to 50Hz at a rate of 1 Hz/s
V2 15	Figure 5	Pre-event voltage at the Grid Entry Point equal to 1p.u. If applicable, solution operating at maximum active power import (demand). Solution operating at zero reactive power output. V5 This step does not need to be completed by 0MW solutions.	Simulate a frequency increase from 50Hz to 52Hz at a rate of 1 Hz/s over a 2 second period. Allow conditions to stabilise for 5 seconds and then decrease the System Frequency from 52Hz to 47Hz at a rate of 1 Hz/s over a 5 second period. Allow conditions to stabilise for 5 seconds, then simulate a frequency increase from 47Hz to 50Hz at a rate of 1 Hz/s

Confidential

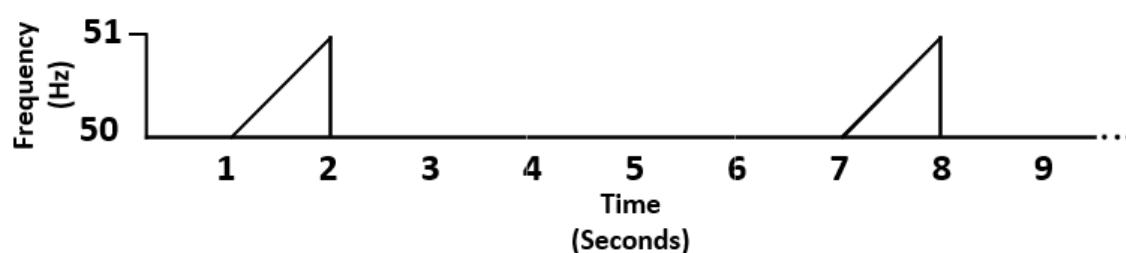


Figure 3: Representation of Step 10

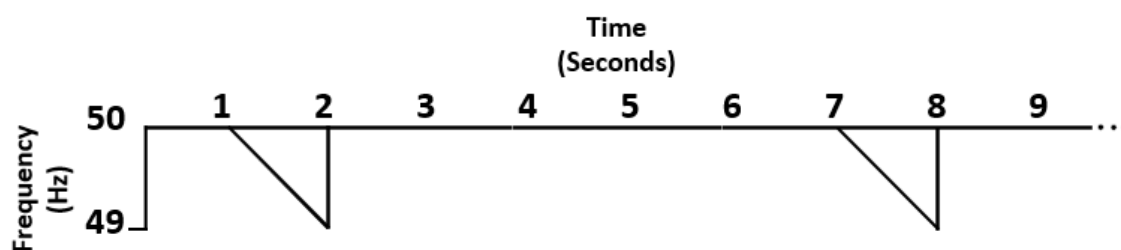


Figure 4: Representation of Step 11

V5 Number of simulations for each solution: 15 for solutions with non-zero MW capability and 9 (Steps 5-12 and 14) for zero MW solutions.

During these simulations the provision of Active Inertia Power must not activate any current limiting functions.

Network Configuration for Frequency and RoCoF tests

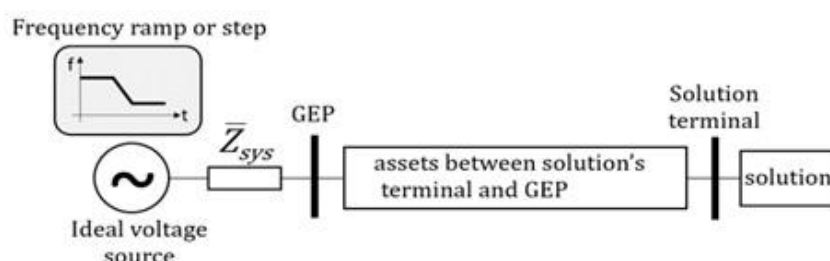


Figure 5: Network configuration for Test 2

Confidential

Figure 5 illustrates the single-line diagram of the system to be simulated in Test 2. The equivalent system impedance (Z_{sys}) has been provided by NESO at tender launch.

The inertia values must also be demonstrated from simulations based on the Active Inertia Power computed for each frequency event.

For all solutions, the inertia value to be used in the tender assessment and contract must be the lowest value demonstrated from Steps 1 to 8.

For all solutions, the inertia values in MVA.s must be computed through Equation 1. For each event, the **Active Inertia Power** (ΔP in Equation 1) must be based on an average across all the recorded samples over the duration of the frequency ramp event.

$$\text{Equation 1: Inertia} = \frac{\Delta P f_0}{2 \times \text{RoCoF}}$$

Where:

ΔP is the **Active Inertia Power** of the **Grid Forming Plant** for a frequency event of 1Hz/s (MW)

RoCoF is the Rate of Change of **Frequency** in Hz/s

f_0 is the pre-fault **System Frequency** (Hz)

For the avoidance of doubt ΔP must be calculated as follows:

$\Delta P = [\text{Average MW provided by the solution at GEP across all recorded samples over the frequency ramp period (1 second)}] - [\text{Initial MW provided by the solution prior to the event}]$.

V2 It is required that an explanation of the plant's performance is provided within the feasibility study report. **V6:** For this, NESO are expecting a full written response from providers, detailing the performance of the plant and any limitations in providing the service.

Confidential

Test 3. Voltage angle change events

The purpose of this test is to understand how a solution will behave under a variety of voltage angle changes at the Grid Entry Point.

V6 The Outcome of these tests must demonstrate

- The performance of a voltage source behind an impedance (ECC.6.3.19)
- For GBGF-I: the performance of the plant when subjected to a Phase Angle Jump Large enough to enter the withstand mode. An explanation of this performance must also be provided.

Failure to demonstrate these requirements may result in the solution failing the Stability Technical Criteria.

The voltage phase jump event must be modelled as a step change in the grid source voltage phase angle. Using other methods such as fault impedance will not be accepted.

V6. Further information around the Phase Angle Jump performance of plant can be found in this [Supplementary Guidance Note](#).

3.1 Test conditions

The Tenderer must note any limitations and observations related to the performance of their solutions.

V2 For all solutions, please refer to Table 4 for a description of the tests required. For solutions which do not have the ability to import or export active power only steps 5, 6, 11, 12, 17, 18, 23 and 24 are required. However, for numbering consistency, please keep numbering the steps as Steps 1-24 even if other steps are not performed.

V6 For all simulations, it should be indicated if the plant is in the withstand mode or not.

V6 For steps 19-24 for GBGF-I plant if the phase angle jump is severe enough to force the plant into the withstand mode an alternative angle can be used that will not force the plant into the withstand mode. This alternative angle should be applied to all steps for 19-24. This should be clearly documented in the report.

Confidential

For each **solution**, the tests described in Table 4 are required:

Table 4: Test 3: Simulations required for each solution.

Step	Network Configuration	Initial Conditions	Simulated Event
1	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power export (generation). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 60 degree drop at the Grid Entry Point.
2	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power export (generation). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 60 degree drop at the Grid Entry Point.
3	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power import (demand). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 60 degree drop at the Grid Entry Point.
4	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power import (demand).	Simulate a 60 degree drop at the Grid Entry Point.

Confidential

		Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	
5	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at Zero active power output. Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 60 degree drop at the Grid Entry Point.
6	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at Zero active power output. Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 60 degree drop at the Grid Entry Point.
V2 7	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power export (generation). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 60 degree rise at the Grid Entry Point.
V2 8	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power export (generation). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 60 degree rise at the Grid Entry Point.

Confidential

V2 9	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power import (demand). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 60 degree rise at the Grid Entry Point.
V2 10	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power import (demand). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 60 degree rise at the Grid Entry Point.
V2 11	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at Zero active power output. Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 60 degree rise at the Grid Entry Point.
V2 12	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at Zero active power output. Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 60 degree rise at the Grid Entry Point.
V2 13	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power export (generation).	Simulate a 30 degree drop at the Grid Entry Point.

Confidential

		Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	
V2 14	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power export (generation). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 30 degree drop at the Grid Entry Point.
V2 15	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power import (demand). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 30 degree drop at the Grid Entry Point.
V2 16	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power import (demand). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 30 degree drop at the Grid Entry Point.
V2 17	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at Zero active power output. Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 30 degree drop at the Grid Entry Point.

Confidential

V2 18	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at Zero active power output. Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 30 degree drop at the Grid Entry Point.
V2 19	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power export (generation). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 5 degree drop at the Grid Entry Point.
V2 20	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power export (generation). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 5 degree drop at the Grid Entry Point.
V2 21	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at maximum active power import (demand). Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 5 degree drop at the Grid Entry Point.
V2 22	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u.	Simulate a 5 degree drop at the Grid Entry Point.

Confidential

		Solution operating at maximum active power import (demand). Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	
V2 23	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at Zero active power output. Solution operating at maximum reactive power export (i.e. lagging mode or reactive power injection).	Simulate a 5 degree drop at the Grid Entry Point.
V2 24	Figure 6	Pre-event voltage at the Grid Entry Point equal to 1p.u. Solution operating at Zero active power output. Solution operating at maximum reactive power import (i.e. leading mode or reactive power absorption).	Simulate a 5 degree drop at the Grid Entry Point.

V2 Number of simulations for each solution: 24 for solutions with non-zero MW capability and 8 for zero-MW solutions.

The Tenderer must demonstrate the performance of their solution(s) under various voltage angle changes. Solutions that do not demonstrate the capability to withstand a voltage phase jump of 60 degrees will not pass the assessment. **V6** Further detail on the assessment for Phase Angle Jumps is detailed below. Solutions must be modelled in detail to capture limitations such as inverter blocking and controller saturation.

Confidential

Network configuration for voltage angle change tests

Figure 6 shows the single-line diagram of the system to be simulated in Test 3. The equivalent system impedance (Z_{sys}) has been provided by NESO at tender launch.

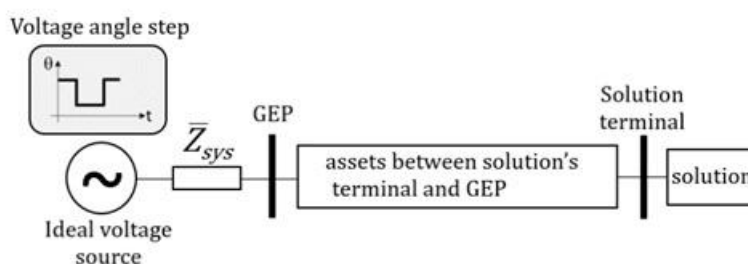


Figure 6: Network configuration for Test 3

V2 The plant must demonstrate the performance of a voltage source behind an impedance for all simulations. **V6** For this, NESO requires a full explanation of the performance of the plant, including any limitations in the plant's performance.

V2: It is required that an explanation of the plant's performance is provided in the feasibility study report. For GBGF-I solutions this must include an explanation of the voltage source behaviour, the implementation of any current limiting functions, and details of the Phase Angle Jump Limit of the plant. For each simulation it is required that an explanation of the performance is provided.

V6: Where any current limiting functions for GBGF-I plant are used, a full explanation of their operation must be provided. This must include as a minimum:

1. Influence on Sequence Components i.e. impact of Current Limiting algorithm used (virtual impedance for example) on the Positive, Negative and Zero Sequence components
2. Influence on Active and Reactive current components
3. If current limitation is achieved through the use of virtual impedance is it bound by time, current or both? How is this activated and deactivated? What is the threshold of the limitation and is this based upon instantaneous or RMS values of current.
4. Following the phase angle jump event how does the plant return to its previous operation point stably?

Confidential

V6: With regards to these tests the following assessments will be made, according to the severity of the angle jump.

For GBGF-I plant NESO will assess the following components:

In the circumstance that the Angle Jump is not severe enough to force the plant into the withstand mode.

- Direction of response for both Current and Power
 - Active current and power showing antiphase behaviour
 - Explanation of the behaviour of reactive and apparent currents provided.
- Reaction Time (5ms or less)
- Following the angle jump, the plant's responses must be well-damped, without sustained or significant oscillations, and shall return stably to the respective pre-disturbance operating point shortly after. The response includes:
 - power responses, including both active and reactive power
 - current responses, including all current components (active and reactive, and both positive- and negative-sequence components).

Any time-series plots and data provided should be of sufficient resolution time-step to enable this performance to be reviewed.

In the circumstance that the Angle Jump is severe enough to force the plant into the withstand mode.

- Direction of response for Current
 - Active current showing inherent initial Antiphase response –in the time before any limiting functions are activated².
 - Explanation of the behaviour of reactive and apparent currents provided.
- Reaction Time (5ms or less)
- Controlled and stable entry and exit into the withstand operating mode and within
- Following the angle jump, the plant's response must be well-damped, without sustained or significant oscillations, and shall return stably to the respective pre-disturbance operating point shortly after. The response includes:
 - power responses, including both active and reactive power
 - current responses, including all current components (active and reactive, and both positive- and negative-sequence components). A full explanation of any current

² If this cannot be demonstrated, an explanation must be provided explaining the performance.

Confidential

limiting function is to be provided in the report to allow NESO to carry out such assessment.

Any time-series plots and data provided must be of sufficient resolution time-step to enable this performance to be reviewed.

For GBGF-S plant NESO will assess the following components:

- Direction of response for both Current and Power
 - Active current and power showing antiphase behaviour
 - Explanation of the behaviour of reactive and apparent current and power provided.
- Reaction Time (5ms or less)
- Following the angle jump, the plant's response must be well-damped, without sustained or significant oscillations, and shall return stably to the respective pre-disturbance operating point shortly after. The response includes:
 - power responses, including both active and reactive power
 - current responses, including all current components (active and reactive, and both positive- and negative-sequence components)

Any time-series plots and data provided must be of sufficient resolution and time-step to enable this performance to be reviewed.